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Final year project Title Submission

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Date:13/06/2018

Project Title 1: AI-Driven Trust and Intelligent Escrow Infrastructure for Informal Digital Commerce

Introduction

Digital commerce in emerging markets is growing rapidly, particularly through informal online platforms and social media channels. However, transactions are often conducted without structured payment protection mechanisms. Buyers transfer funds directly to sellers without guarantees of delivery, creating significant risks of fraud, disputes, and financial loss.

This project proposes the development of an AI-Driven Trust and Intelligent Escrow Infrastructure, a web-based platform that acts as a secure intermediary between buyers and sellers. In addition to holding payments in escrow, the system integrates an AI-based trust scoring engine that evaluates transaction behavior, detects anomalies, and assesses risk before funds are released. The platform combines escrow management, intelligent monitoring, and dispute resolution to create a secure digital commerce environment.

Problem Statement

Current informal e-commerce systems lack intelligent trust evaluation and structured payment mediation. Transactions occur without predictive risk analysis, increasing exposure to fraud and delivery disputes. A secure and AI-supported escrow infrastructure is required to strengthen trust and reduce transactional uncertainty in digital marketplaces.

Objectives

General Objective:

To design and develop an AI-driven escrow infrastructure that enhances transaction security and trust in informal digital commerce.

Specific Objectives:

- Develop secure user management for buyers, sellers, and administrators.
- Implement an escrow wallet mechanism to temporarily hold transaction funds.

- Design a transaction lifecycle model with defined states (Initiated, Funded, Shipped, Completed, Disputed).
- Integrate an AI-based trust scoring and anomaly detection module.
- Provide structured dispute resolution and administrative decision management.

Example Scenario

A buyer initiates a transaction and deposits payment into escrow. The AI module evaluates transaction history and behavioral indicators to assign a trust score. The seller ships the product, and upon delivery confirmation, the system releases funds automatically. If abnormal behavior is detected or a dispute is filed, the system flags the case for administrative review before releasing payment.

Conclusion

The proposed infrastructure enhances digital trust by combining secure escrow mechanisms with intelligent risk assessment. It reduces fraud, formalizes online transactions, and strengthens confidence in emerging digital commerce ecosystems.

Project Title 2: AI-Enhanced Adaptive Enterprise Application Platform with Dynamic Schema and Workflow Generation

Introduction

Organizations frequently encounter limitations when using rigid enterprise software systems that do not adapt to evolving operational processes. Custom software development is expensive and time-consuming, while off-the-shelf solutions often lack flexibility.

This project proposes an AI-Enhanced Adaptive Enterprise Application Platform capable of dynamically generating database schemas, workflows, and business logic through metadata configuration. The platform enables organizations to define and modify their processes without extensive code changes. An AI-assisted optimization component analyzes workflow execution patterns to recommend performance improvements and structural refinements.

Problem Statement

Traditional enterprise systems are static and difficult to modify once deployed. Organizations require adaptable platforms that can evolve alongside changing business requirements. A configurable and intelligent enterprise architecture is needed to minimize redevelopment costs while improving operational efficiency.

Objectives

General Objective:

To design and develop an adaptive enterprise platform that dynamically generates workflows and data structures while incorporating AI-assisted optimization.

Specific Objectives:

- Implement dynamic schema and entity generation based on metadata definitions.
- Enable configurable workflow modeling and process automation.
- Integrate AI-based analysis for workflow optimization and bottleneck detection.
- Support multi-tenant architecture with secure data isolation.
- Provide role-based access control and modular system components.

Example Scenario

An organization defines its operational processes through configuration tools within the platform. The system automatically generates corresponding database structures and workflows. As users interact with the system, the AI module analyzes execution patterns and suggests improvements to enhance efficiency and reduce processing delays.

Conclusion

The platform provides a flexible and intelligent foundation for enterprise systems by combining dynamic configuration with AI-assisted optimization. It reduces development redundancy and enables organizations to adapt their digital infrastructure efficiently.

Project Title 3: AI-Enabled Event-Driven Microservices Architecture for Scalable Distributed Systems

Introduction

Modern applications require scalable and resilient architectures capable of handling distributed workloads. Monolithic system designs often limit scalability, fault isolation, and independent deployment. Event-driven microservices architectures address these limitations by decomposing systems into independent services communicating through asynchronous messaging.

This project proposes the design and implementation of an AI-Enabled Event-Driven Microservices Architecture. The system integrates intelligent monitoring mechanisms that analyze service performance metrics and detect anomalies in real time. By combining distributed service communication with AI-assisted monitoring, the architecture aims to enhance scalability, reliability, and fault tolerance.

Problem Statement

Traditional application architectures struggle with scalability and resilience under dynamic workloads. While microservices improve modularity, they introduce complexity in monitoring and failure detection. An intelligent monitoring and anomaly detection mechanism is required to enhance reliability and system stability in distributed environments.

Objectives

General Objective:

To design and implement an AI-enabled event-driven microservices architecture that improves scalability, resilience, and intelligent monitoring in distributed systems.

Specific Objectives:

- Develop independent microservices communicating through an event broker.
- Implement service discovery and load balancing mechanisms.

- Establish centralized logging and performance monitoring.
- Integrate AI-based anomaly detection for identifying system irregularities.
- Simulate fault isolation and recovery scenarios to evaluate resilience.

Example Scenario

Multiple microservices process user requests asynchronously through an event broker. The AI monitoring module continuously analyzes latency, error rates, and throughput metrics. When abnormal patterns are detected, the system alerts administrators or triggers predefined recovery mechanisms to prevent service disruption.

Conclusion

The proposed architecture demonstrates how intelligent monitoring can enhance distributed microservices systems. By integrating AI-driven anomaly detection with event-based communication, the system improves scalability, reliability, and operational efficiency.